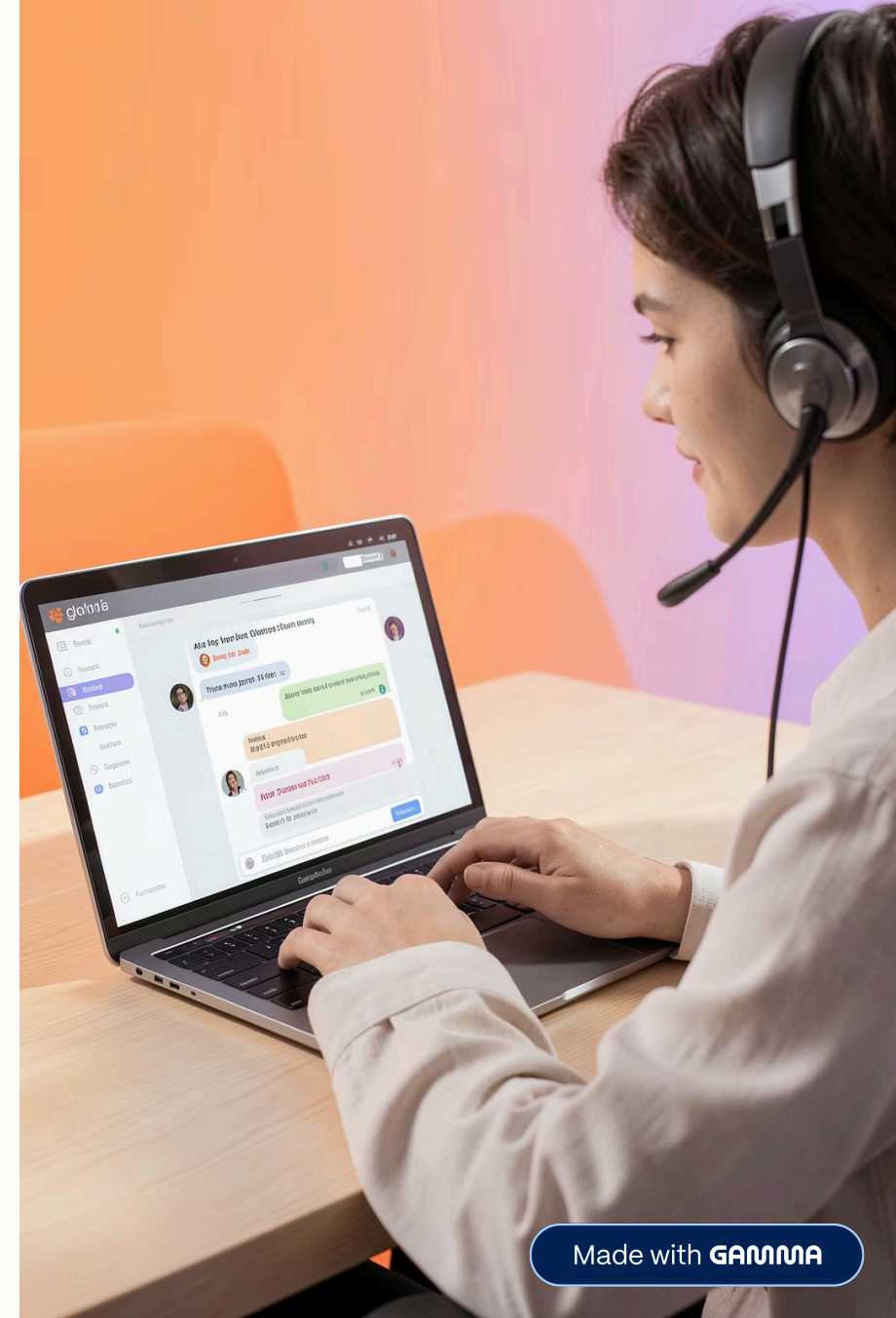


Atlas : Real-time Sentiment Analysis Agent

Customer Support Automation

DATA 298B — Team 1 | San José State University

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The Problem We Solved

Slow Responses

Customers frustrated by generic replies with no order context

Overwhelmed Agents

Repetitive, predictable queries consuming human support capacity

No Escalation

No real-time routing for frustrated or high-urgency customers

Isolated Channels

Web and email handled separately with no unified pipeline



Project Requirements

Goal

NLP-driven agentic chatbot with **real-time sentiment analysis**, intelligent escalation, and multi-channel support.

Technologies

Python · HuggingFace Transformers · spaCy · Docker

Outcome

Intelligent chatbot automating customer support and enhancing satisfaction

Key FeaturesA



Auto-Escalation

Routes to human agent when frustration threshold is reached



Identity Gate

Email + order ID match required before exposing personal data



8 Integrated Tools

Order lookup · refund · cancel · product search · policy search · escalation · email · account info



Dual RAG Pipeline

1.4M+ Amazon product catalog + support policies knowledge base



Multi-channel

Production web UI + inbound email webhook via full Docker stack

Data Sets

Training Data

NebulaBytes and **ByteText** were combined to train the model.

This dataset was used to teach the system core response patterns, intent handling, and conversational behavior.

Product & Customer Data

Amazon product dataset was used for product details.

customer details were added to simulate real-world support scenarios and test end-to-end interactions.

Model Selection Journey

1

LLaMA-3.2-1B

Lost multi-turn context; responses too generic for support scenarios

2

SmolLM2-1.7B

Better instructions but weak on policy knowledge retrieval

3

Qwen2.5-1.5B

Solid general performance; inconsistent tone and empathy

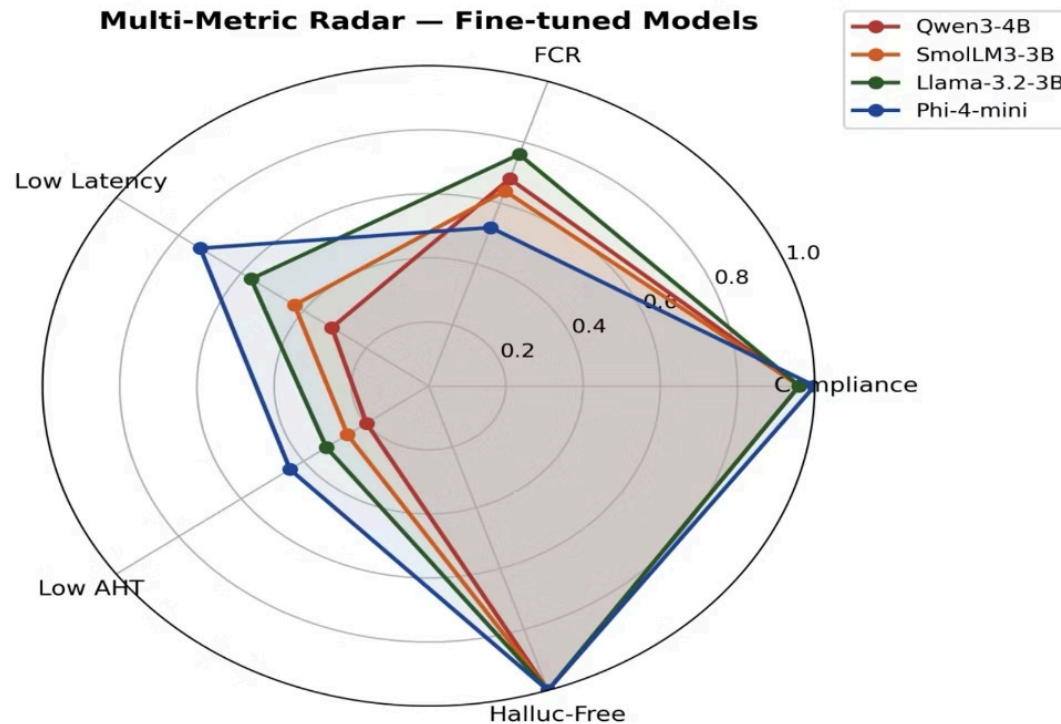
4

Phi-4-mini (3.8B) ✓

Best instruction following · coherent multi-turn · policy-grounded

Why Phi-4-mini?

Why Phi-4-mini Wins:



→ Best Instruction-Following

Top performer among all evaluated compact models

→ Multi-turn Context

Maintains order context across turns without repetition

→ Structured & Empathetic

Produces professional, policy-grounded support responses

→ Reasoning-Optimized

Designed for reasoning tasks — critical for policy lookups and order workflows. Fine-tuned via PEFT on A10G GPU.

System Architecture



Sentiment and intent detection runs sub-millisecond via keyword/regex. The deterministic tool router eliminates LLM guessing, while a 5-layer safety net handles stalls and data checks.

Technical Stack

LLM

Phi-4-mini QLoRA fine-tuned · PEFT adapter · HuggingFace Endpoint

Backend

FastAPI + Python · PostgreSQL 16 · Docker Compose 4-service stack

Vector DB

Qdrant v1.12.4 cosine similarity · all-MiniLM-L6-v2 384-dim embeddings

Email

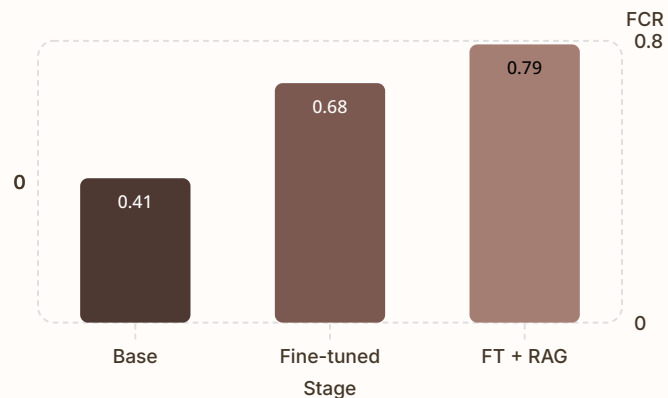
SMTP + MailHog (dev) · inbound webhook + outbound auto-email

Evaluation

G-Eval via Gemini 2.5 Flash · ROUGE-L · Task Success Rate

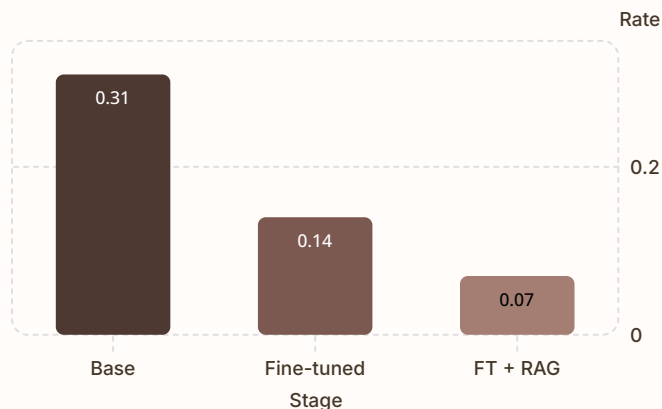
Detailed Evaluation Metrics

First Contact Resolution (FCR)



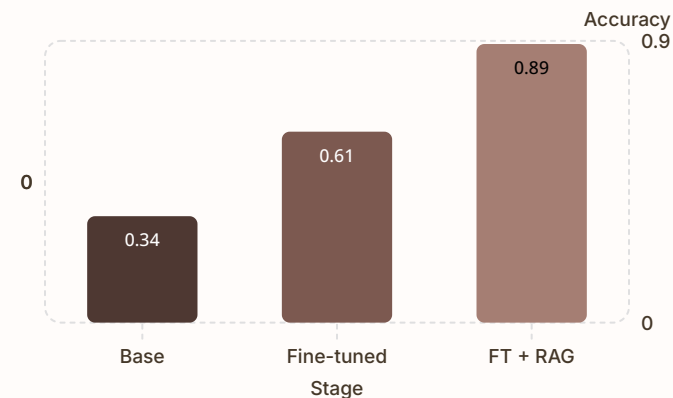
Improvements of **+0.27** (Fine-tuned) and **+0.11** (FT + RAG).

Hallucination Rate



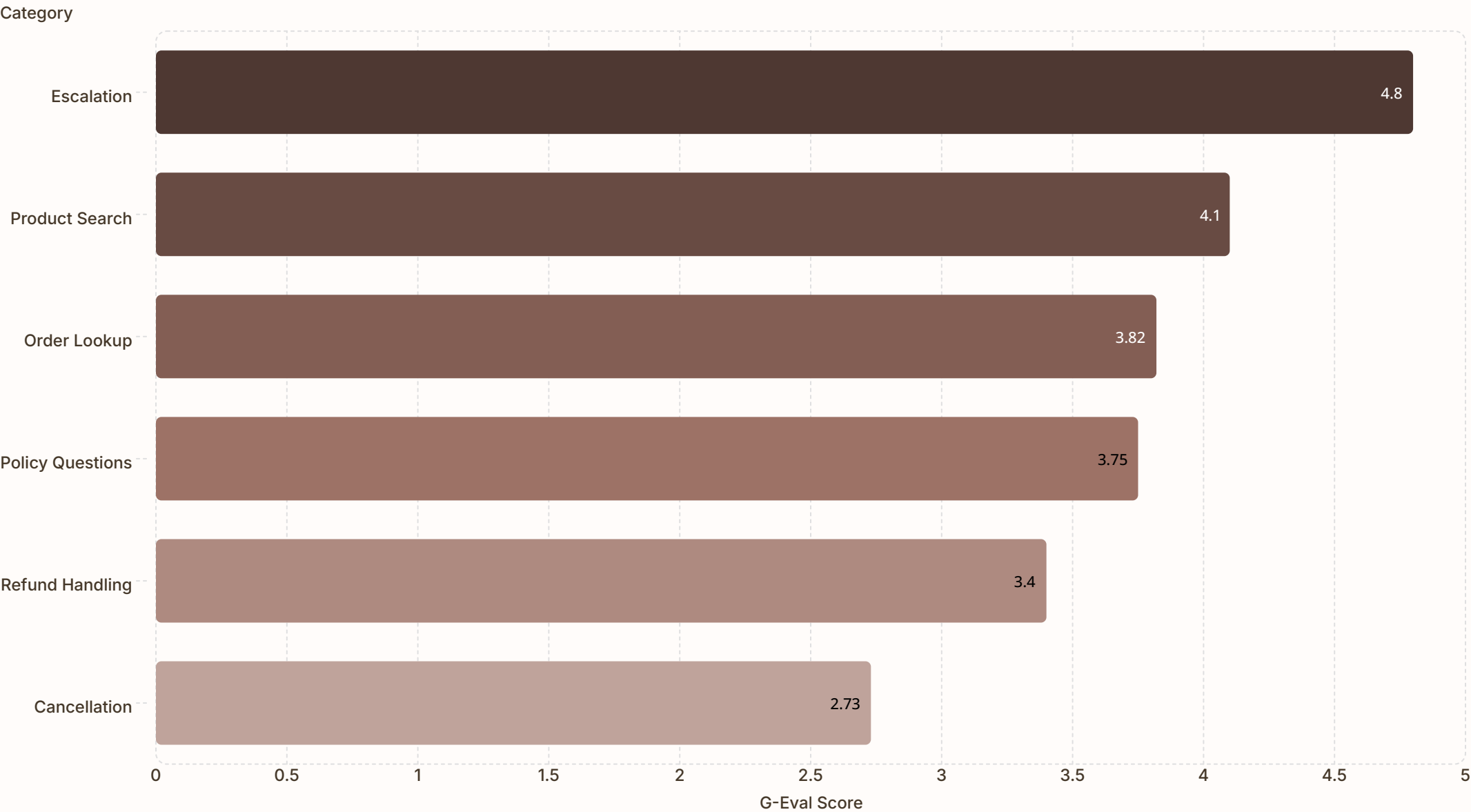
Reduction of **-0.17** (Fine-tuned) and **-0.07** (FT + RAG).

Policy Answer Accuracy



Improvements of **+0.27** (Fine-tuned) and **+0.28** (FT + RAG).

Evaluation Results



250 Test Cases

72.3%

Task Success Rate

186 of 250 cases fully resolved

3.79

G-Eval Average

Out of 5.0 across all dimensions

Evaluation dimensions: Relevance · Faithfulness · Completeness · Tone · Groundedness

Production-Ready Deliverables

Live Web UI Session list · quick-action chips · tool invocation transparency panel	REST API /chat · /health · /rag/search · /policy/search · /webhooks/email/inbound
Beyond Scope Identity gate · Dual RAG · Auto-escalation · 8 integrated tools · Policy KB	One-Command Deploy docker compose up — fully self-contained · zero critical bugs

Team 1 — DATA 298B | San José State University · Instructor: Prof. Simon S. Shim, Ph.D.